

Application of AI-Assisted Adaptive Learning System in Special Education

Yongji Shen and Kexiong Cui(✉)

Sanya Aviation and Tourism College, Sanya 572000, Hainan, China
kx_cui1996@sohu.com

Abstract. Many children are facing the challenge of being marginalized by mainstream education systems due to disabilities such as visual, auditory, intellectual developmental, or autism. They often have difficulty accessing educational resources that meet their special needs, which hinders their comprehensive development and integration into society. However, with the rapid advancement of technology, artificial intelligence (AI) technology has brought new solutions to such problems. In the field of education, adaptive learning systems, as an important branch of AI applications, are gradually showing great potential in special education. This article proposes an innovative AI adaptive learning system that deeply integrates deep learning (DL) algorithms. It can intelligently analyze students' learning habits, interests, and ability status, and dynamically adjust teaching content, difficulty, and progress based on this, designing personalized learning plans for special needs children with different needs. Experimental results have shown that the system can accurately recommend the educational resources that students need, thereby significantly increasing their interest and participation in learning.

Keywords: Artificial intelligence · Adaptive learning system · Special education

1 Introduction

In the information wave of the 21st century, the rapid development of Internet technology not only reshapes the appearance of traditional industries, but also profoundly affects the field of education, especially the field of special education [1]. For special needs children, this change is particularly significant as they often face more complex and diverse learning needs, and traditional education models often struggle to provide sufficient flexibility and personalized support [2]. The introduction of AI technology has brought unprecedented development opportunities for special education, opening up a path towards personalized, efficient, and adaptable learning for these special needs children [3]. Personalized learning, as a key path to achieving comprehensive student development, emphasizes tailoring learning content and teaching strategies based on individual differences of students to meet their unique learning needs [4]. In the field of special education, this concept is particularly important. Due to physiological, psychological, or cognitive differences, special needs children have more diverse and complex educational needs. With the increasing emphasis on special education in society, special education schools are gradually making personalized learning the core goal of school construction and development [5].

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025
M. Virvou et al. (Eds.): DCSL 2024, LAIS 54, pp. 234–242, 2025.
https://doi.org/10.1007/978-3-031-98607-9_23

This not only requires schools to have the ability to identify individual differences among students, but also to build a support system that can dynamically adjust teaching content, methods, and progress. Against the backdrop of rapid development in mobile internet and smart education, personalized adaptive learning systems have become a research hotspot in the field of educational technology [6]. This type of system is based on big data and can intelligently identify learners' learning patterns and preferences by continuously collecting and analyzing their learning data. It can then dynamically adjust learning resources, difficulty, and progress to achieve optimal learning outcomes [7]. For example, Marino et al. [8] proposed a DL based personalized recommendation model for digital library resources, which mines user behavior patterns through DL algorithms and provides learners with accurate learning resource recommendations. Abu et al. [9] utilized learners' social network information for collaborative filtering and recommendation, predicting trust values and ratings in social relationships to recommend learning resources to new learners. This method is particularly valuable in special education environments because it can fully utilize peer influence and promote the construction of learning communities.

The joint neural collaborative filtering recommendation system proposed by Xavier et al. [10] further improves the accuracy and personalization of recommendations by combining deep feature learning, deep interaction modeling, and association matrix. Marino et al. [11] used GPS technology to recommend English learning resources, effectively improving learners' English oral communication skills. These studies not only promote the development of personalized recommendation technology, but also provide valuable references for adaptive learning systems in the field of special education. This article proposes an innovative AI adaptive learning system that deeply integrates DL algorithms, aiming to provide more refined and intelligent learning support tailored to the learning characteristics of special needs children. By intelligently analyzing students' learning habits, interests, and abilities, the system can dynamically adjust teaching content, difficulty, and progress to ensure that each student can learn in the most suitable environment for themselves.

2 Methodology

2.1 System Construction

In today's increasingly popular digital education, personalized adaptive learning systems have become a key tool for improving learning efficiency and effectiveness. This type of system dynamically adjusts the recommendation of learning resources and the planning of learning paths by deeply understanding the individual characteristics, interest preferences, learning styles, and ability levels of learners, aiming to provide tailored learning experiences for each learner [12]. The starting point of personalized adaptive learning systems lies in the precise analysis of learner characteristics. This includes but is not limited to learners' mastery of basic knowledge, learning speed, preference types, emotional states, and learning motivation [13]. Traditional methods may rely on questionnaire surveys or teacher evaluations, but modern systems make more use of big data analysis and machine learning (ML) algorithms to automatically capture

and analyze learners' online learning behavior data, thus more comprehensively characterizing learners' profiles. Learning content recommendation is the core function of adaptive learning systems, which directly determines whether learners can obtain the most suitable learning materials for themselves.

Traditional recommendation algorithms, such as Probability Matrix Factorization (PMF), can effectively utilize user item evaluation matrices for prediction, but ignore rich edge information such as users' social network relationships, item label descriptions, and users' historical behavior patterns, which are crucial for improving recommendation accuracy. To overcome this limitation, this paper proposes a new algorithm that combines DL and incorporates edge information. Specifically, we use a fourth-order convolutional neural network (CNN) to process and analyze user and resource profile text information, to extract deep level feature representations. Using a pre-trained word2vec model, convert text information into dense representations in a high-dimensional vector space, providing input for subsequent convolution operations. In the second layer of CNN, multiple convolution kernels with different window sizes are applied to extract local features from the text vector. These convolution kernels share parameters, reducing model complexity and enhancing generalization ability. The third layer downsamples the feature map output by the convolutional layer through pooling operation, preserving the most important feature information while reducing computational complexity.

Finally, in the fourth layer, the pooled feature vectors are transformed into fixed length text representation vectors using a fully connected layer. These vectors not only contain deep level information about the text content, but also imply unique features of users or resources. The improved structure of the CNN-PMF model is shown in Fig. 1. Based on the analysis of learner characteristics and content recommendations, the system needs to further plan personalized learning paths. This involves serializing and arranging learning resources to ensure that the difficulty of the learning content gradually increases, while also conforming to the cognitive patterns and interests of learners. Learning effectiveness evaluation is an indispensable part of adaptive learning systems, which not only helps learners understand their own progress, but also provides a basis for adjusting recommendation strategies for the system. By continuously collecting data during the learning process, such as test scores, participation, feedback, etc., combined with ML models for multidimensional analysis, learning effectiveness can be accurately evaluated.

2.2 Algorithm Principle

The PMF model is a leader in collaborative filtering algorithms, known for its representativeness and wide applicability. The core principle of this model is to fill in the unknown rating values by decomposing and reconstructing the rating matrix. Specifically, this process first involves constructing a "user item" rating matrix R, which is then decomposed into the product of two low dimensional matrices U, V. By utilizing the inner product of U, V, we can reconstruct a new rating matrix $\hat{R}$. The beauty of this step is that it allows items in the original rating matrix R that have not been rated to also receive corresponding predicted ratings. On this basis, we exclude items that have already been rated by users, and only sort the remaining items based on the "reconstructed" rating values to generate the final item recommendation list. The objective function

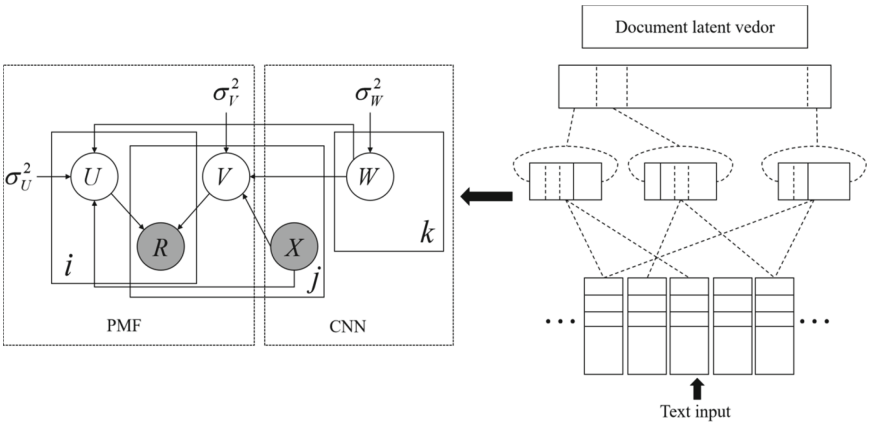


Fig. 1. Structure of CNN-PMF model

of this model is to:

$$Loss = \sum_{i=1}^{N}\sum_{j=1}^{M} I_{ij}\left(R_{ij} - U_i^T V_j\right)_2 + \lambda_U \sum_{i=1}^{n} \|U\|_2 + \lambda_V \sum_{j=1}^{m} \|V\|_2 \tag{1}$$

Among them, R_{ij} represents the actual observed score; And $U_i^T V_j$ represents the rating predicted by the model. In order to avoid overfitting, regularization parameters λ_U, λ_V are introduced, which are set as $\lambda_U = \sigma^2/\sigma_U^2, \lambda_V = \sigma^2/\sigma_V^2$, where $\sigma^2, \sigma_U^2; \sigma^2, \sigma_V^2$ represents different variance terms. In addition, n, m refers to the specific values of the total number of users and the total number of projects, that is, there are n users and m projects. I_{ij}, as an indicator function, takes a value of 1 in the presence of ratings and 0 in the absence of ratings.

The process of solving the parameter matrix U, V is transformed into an optimization problem, as described below.

$$U, V = \arg\min_{U,V} J(U, V) \tag{2}$$

Usually, we use batch gradient descent (BGD) to solve the above equation. BGD uses an iterative optimization strategy to continuously update the elements U_{im}, V_{mj} in the parameter matrix U, V, aiming to gradually approximate the global or local minimum of the loss function $J(U, V)$. The specific iterative update rules follow the following principles:

$$U_{im}^{n+1} := U_{im}^{n} - \alpha \cdot \frac{\partial}{\partial U_{im}} J(U, V) \tag{3}$$

$$V_{mj}^{n+1} := V_{mj}^{n} - \alpha \cdot \frac{\partial}{\partial V_{mj}} J(U, V) \tag{4}$$

In the formula, n represents the number of iterations; α is the learning rate.

Although CNN models have limitations in mining key and representative information in project review documents, they are able to capture global information as well as correlations between contexts. Through the multi-layer convolutional structure in the CNN model, we can effectively extract the interrelationships between words in project comment documents, learn the global features of the project and the connections between context, and generate implicit representations of the project. In the convolutional layer, the model extracts feature from the word vector matrix G, using sliding windows of sizes 3, 4, and 5 for convolution operations to capture different text features. This convolution process can be formally expressed as the following formula:

$$A = relu\left(\sum_{i=0}^{n}\sum_{j=0}^{m} w_{i,j}G\right) \tag{5}$$

Among them, A represents the activation value calculated on a certain convolution kernel; $w_{i,j}$ represents the weight parameters in the convolutional kernel; *relu* is the activation function selected in this article to introduce nonlinear characteristics; G represents the input of the convolutional layer, which is the word vector matrix. Through this convolution operation, the model is able to extract feature information from the input text.

In the output layer stage, the new feature quantities are further mapped to generate the final latent feature representation of the project. Specifically, CNN is used to convert raw project comment documents into project feature vectors, and then output a deep semantic representation matrix. This representation matrix is then mapped to a L dimensional space to capture the core features of the item, and the mapping process can be expressed as the following mathematical formula:

$$cnn\left(w', Y_j\right) = relu(h_2\{relu(h_1 d_z + b_1)\} + b_2) \tag{6}$$

Among them, h_1, h_2 represents the mapping matrix, and b_1, b_2 is the corresponding bias term; d_z represents the output of the pooling layer; Y_j is the input data for CNN; w' covers all parameters of CNN. Ultimately, the output dimension of CNN needs to be consistent with the hidden feature vector dimension in the PMF model to ensure compatibility between the two in subsequent processing.

3 Result Analysis and Discussion

To fully demonstrate the performance advantages of the system proposed in this article, we have specially planned a set of comparative experiments to directly compare the performance of our system with traditional deep neural network (DNN) systems in educational resource recommendation. Table 1 provides a detailed comparison of the recommendation accuracy between our system and traditional systems. It can be clearly seen from the table that the accuracy achieved by the system in this article is much higher than that of traditional systems. This significant advantage is mainly attributed to the CNN-PMF model adopted by the system in this article. This model innovatively combines the advantages of CNN and PMF, leveraging both CNN's outstanding ability in

text feature extraction and PMF's unique role in collaborative filtering recommendation. Specifically, the CNN component is responsible for in-depth analysis of project comment documents, extracting key information and contextual connections, and generating high-quality project feature vectors; The PMF component accurately predicts users' potential interest in the project by decomposing the rating matrix. These two parts work together to improve the accuracy and personalization level of the recommendation system.

Table 1. Accuracy Comparison /%

Number of tasks/pieces	This article system	Traditional system
10	84.7	74.5
20	85.6	75.2
50	86.8	76.4
80	87.3	77.1
100	88.2	78.5

Table 2 provides a detailed comparison of the performance of our system and traditional DNN system in terms of student education resource recommendation recall rate. Recall rate, as another key indicator for evaluating the effectiveness of recommendation systems, measures the proportion of relevant resources successfully recommended by the system among all relevant resources. The intuitive data display shows that the recall rate achieved by the system in this article far exceeds that of traditional systems, which strongly proves the outstanding performance of the system in the field of educational resource recommendation. This significant achievement is mainly attributed to the CNN-PMF model adopted by the system in this article. This model cleverly integrates the technical essence of CNN and PMF, not only improving the accuracy of recommendations, but also greatly enhancing the breadth of recommendations. Specifically, the CNN component, with its excellent text feature extraction ability, deeply analyzes project review documents and extracts highly summarized project feature vectors. The PMF component, on the other hand, accurately grasps users' potential interests in the project by parsing the rating matrix, providing users with more personalized recommendations that are more tailored to their needs.

Figure 2 visually compares the difference in time consumption between the system proposed in this article and the traditional DNN system for student educational resource recommendation tasks. From the data in the figure, it can be seen that compared to traditional systems, the system in this paper significantly reduces the time consumption when performing the same recommendation task, highlighting the efficiency of the system in educational resource recommendation. The efficiency behind this is mainly attributed to the CNN-PMF model adopted by the system in this article. This model combines the technical advantages of CNN and PMF, not only improving the accuracy and comprehensiveness of recommendations, but also significantly accelerating the processing speed of the system. Specifically, the CNN part which has strong feature extraction capability can extract key information from project review documents quickly, generate high-quality

Table 2. Comparison of Recall Rates /%

Number of tasks/pieces	This article system	Traditional system
10	78.3	64.7
20	78.7	65.3
50	79.2	67.4
80	80.4	68.1
100	80.7	69.2

project feature vectors and shorten the time of feature extraction effectively. This is complemented by the PMF, which has the task of predicting quickly economical-use matrix factorization algorithm users' interest in the project, speeding up the recommendation.

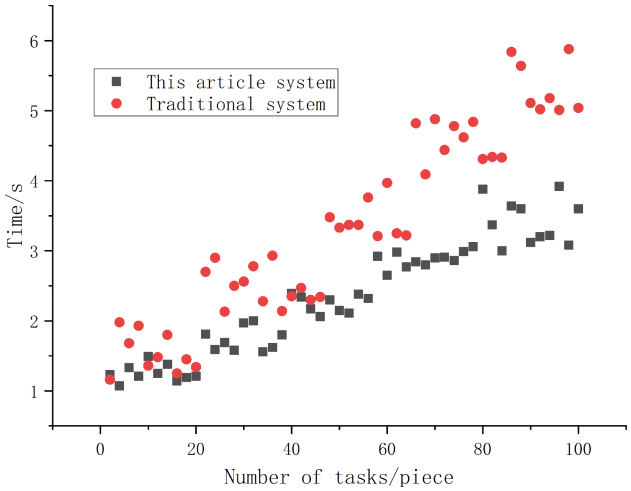


Fig. 2. Time comparison

Figure 3 visually compares the performance of our system with traditional DNN systems in terms of F1 score for student educational resource recommendation tasks. The F1 score, as a key indicator for evaluating the comprehensive performance of recommendation systems, integrates precision and recall, comprehensively revealing the system's overall performance in recommendation tasks. The data in the figure shows that compared to traditional systems, the F1 value of our system is significantly higher, which strongly proves the excellent comprehensive performance of new system in educational resource recommendation. Central to this performance is the CNN-PMF model employed by the system described. The model ingeniously leverages the technical benefits of CNN and PMF, which not only increases the accuracy and completeness of recommendations but also boosts the entire system performance. Due to the competent

feature extraction function of CNN components, it would be able to extract key information from project review documents, so as to mine to get project high-definition features vectors, thus providing a strong basis for accurate recommendation. On the other hand, the PMF component also does well in capturing the user's potential interests in the project using efficient matrix factorization algorithms, with a promotion of the recall rate and the overall improvement of the recommendation system's performance.

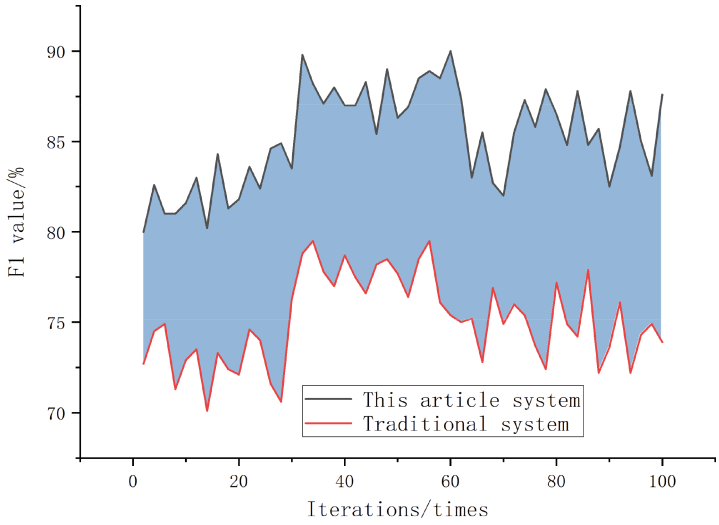


Fig. 3. Comparison of F1 values

4 Conclusion

The research findings of this article introduce an innovative AI adaptive learning system that closely integrates DL algorithms, demonstrating the extensive potential of AI in the field of education. This system can intelligently analyze students' learning habits, interests, and ability levels, and flexibly adjust teaching content, difficulty, and progress, accordingly, providing personalized learning plans for special needs children. This innovative measure not only meets the diverse learning needs of students, but also greatly stimulates their interest and participation in learning, opening up new avenues for improving educational equity and quality. The experimental results show that this system exhibits high accuracy in recommending educational resources, effectively meeting the actual learning needs of students and promoting the improvement of learning outcomes. However, while acknowledging the achievements of the system, we should also acknowledge its limitations. In the face of complex or special learning needs, the adaptive flexibility and accuracy of the system still needs to be improved; At the same time, the scalability and compatibility of the system are also key technological challenges that need to be addressed in the future to ensure that it benefits a wider range of students from different backgrounds.

References

1. Marras, M., Boratto, L., Ramos, G., Fenu, G.: Equality of learning opportunity via individual fairness in personalized recommendations. Int. J. Artif. Intell. Educ. **32**, 636–684 (2022)
2. Li, Q.: The use of artificial intelligence combined with cloud computing in the design of education information management platform. Int. J. Emerg. Technol. Learn. (iJET) **16**, 32–44 (2021)
3. Prinsloo, P., Khalil, M., Slade, S.: Vulnerable student digital well-being in AI-powered educational decision support systems (AI-EDSS) in higher education. Br. J. Educ. Technol. **55**, 2075–2092 (2024)
4. Lu, Y., Wang, D., Chen, P., Meng, Q., Yu, S.: Interpreting deep learning models for knowledge tracing. Int. J. Artif. Intell. Educ. **33**, 519–542 (2023)
5. Lodge, J.M., Thompson, K., Corrin, L.: Mapping out a research agenda for generative artificial intelligence in tertiary education. Aust. J. Educ. Technol. **39**, 1–8 (2023)
6. Hu, S.: The effect of artificial intelligence-assisted personalized learning on student learning outcomes: a meta-analysis based on 31 empirical research papers. Sci. Insights Educ. Front. **24**, 3873–3894 (2024)
7. Jin, D., Wen, X., Wen, Y.: Personalized learning efficiency data analysis based on multi-scale convolution architecture and hybrid loss. Neural Comput. Appl. **36**, 9753–9766 (2024)
8. Marino, M.T., Vasquez, E., Dieker, L., Basham, J., Blackorby, J.: The future of artificial intelligence in special education technology. J. Spec. Educ. Technol. **38**, 404–416 (2023)
9. Abu Mukh, Y.N., Abd-Rabo, A.M., Tarteer, S.: Computer competencies needed for implementing artificial intelligence in special education schools from the perspective of pre-service teachers. Trends Higher Educ. **3**, 602–622 (2024)
10. Xavier, J.L., Khoury, J., Phen, H.M., Fragomen, A.T., Rozbruch, S.R., Kahn, M.: Evaluating the efficacy of natural language processing artificial intelligence models as a patient education tool for stature lengthening surgery and reconstruction. J. Limb Lengthening Reconstr. **10**, 22–27 (2024)
11. Marino, M.T., Vasquez, E., III.: Special education administrators use of artificial intelligence (AI) to synthesize data. J. Special Educ. Leadersh. **37**, 62–76 (2024)
12. Smith, S.J., Rowland, A., Goldman, S., Carreon, A.: A guide for special education leaders to utilize artificial intelligence: students' perspectives for future consideration. J. Special Educ. Leadersh. **37**, 77–92 (2024)
13. Malenica, I., Phillips, R.V., Chambaz, A., Hubbard, A.E., Pirracchio, R., van der Laan, M.J.: Personalized online ensemble machine learning with applications for dynamic data streams. Stat. Med. **42**, 1013–1044 (2023)